

Question

If the atomic weight of an element is 23 times that of the lightest element and it has 11 protons, then it contains

- A** 11 protons, 23 neutrons, 11 electrons
- B** 11 protons, 11 neutrons, 11 electrons
- C** 11 protons, 12 neutrons, 11 electrons
- D** 11 protons, 11 neutrons, 23 electrons

Question

The number of protons, neutrons and electrons in $^{175}\text{Lu}^{71}$ respectively, are:
[NEET-2020]

- A** 71, 104 and 71
- B** 104, 71 and 71
- C** 71, 71 and 104
- D** 175, 104 and 71

Question

Chlorine atom differs from chloride ions in the number of

- A** Proton
- B** Neutron
- C** Electrons
- D** Protons and electrons

Question

The number of neutrons in tripositive aluminum ion with mass number 27 is

- A** 14
- B** 16
- C** 18
- D** 20

Question

Atomic number and mass number of an element M are 25 and 52 respectively. The number of electrons, protons and neutrons in M^{2+} ion are respectively.

- A** 25, 25 and 27
- B** 25, 27 and 25
- C** 27, 25 and 27
- D** 23, 25 and 27

Question

Atomic weight of chlorine is 35.5. It has two isotopes of atomic weight 35 and 37. What is the percentage of the heavier isotope in the sample?

- A** 5
- B** 10
- C** 25
- D** 20

Question

B has two isotopes ^{10}B (19%) and ^{11}B (81%). The atomic mass of B is.

- A** 10.81
- B** 11.5
- C** 11
- D** 10.5

Question

Average atomic mass of magnesium is 24.31 amu. This magnesium is composed of 79% of ^{24}Mg and remaining 21% of ^{25}Mg and ^{26}Mg . Calculate % of ^{26}Mg .

A 10

B 11

C 15

D 16

Question

Select the correct Statements from the following:

- A. Atoms of all elements are composed of two fundamental particles.**
- B. The mass of the electron is 9.10939×10^{-31} kg.**
- C. All the isotopes of a given elements show same chemical properties.**
- D. Protons and electron are collectively known as nucleons.**
- E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.**

Choose the correct answer form the options give below:

[NEET-2023]

A C, D and E only

B A and E only

C B, C and E only

D A, B and C only

Question

Chemistry was not studied for its own sake, rather it came up as a result of search for

- A** Philosopher's stone
- B** Elixir of life
- C** Paras
- D** All of the above

Question

Which of the following is not described in Charaka Samhita?

- A** Preparation of nitric acid and oxides of Cu, Sn and Zn.
- B** Use of various parts of plant for the preparation of Asavas.
- C** Preparation of perfumes and cosmetics.
- D** Use of Bhasma of metals for treatment of ailments.

Question

Who among the following was the first proponent of atomic theory?

- A** Acharya Kanda
- B** Chakrapani
- C** Nagarjuna
- D** None of the above

Question

The two effective drugs which act as life saving drugs for cancer therapy and AIDS victims respectively are.

- A** AZT (azidothymidine) and taxol.
- B** taxol and cisplatin.
- C** taxol and AZT (azidothymidine).
- D** AZT (azidothymidine) and cisplatin.

Question

Which of the following environmentally hazardous refrigerant responsible for ozone depletion in the stratosphere?

- A** CFCs (Chlorofluorocarbons)
- B** CFM (Chlorofluoromethane)
- C** CC (Chloriodocarbons)
- D** CIM (Chloriodomethane)

Question

Choose the Incorrect Statement about Dalton's Atomic Theory

[04 April, 2024 (Shift-II)]

- A** Compounds are formed when atoms of different elements combine in any ratio
- B** All the atoms of a given element have identical properties including identical mass
- C** Matter consists of indivisible atoms
- D** Chemical reactions involve reorganization of atoms

Question

The incorrect postulates of the Dalton's atomic theory are:

(A) Atoms of different elements differ in mass.

(B) Matter consists of divisible atoms.

(C) Compounds are formed when atoms of different element combine in a fixed ratio.

(D) All the atoms of given element have different properties including mass.

(E) Chemical reactions involve reorganisation of atoms.

Choose the correct answer from the options given below:

[05 April, 2024 (Shift-I)]

A (B), (D), (E) only

B (B), (D) only

C (A), (B), (D) only

D (C), (D), (E) only

Question

Assertion: Solids have definite volume and shape.

Reason: In solids, the constituent particles are very close to each other and there is not much freedom of movement.

- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

Question

Assertion: On heating, a solid usually changes to a liquid and the liquid on further heating changes to the gaseous state.

Reason: Arrangement of constituent particles is different in solid, liquid and gaseous state.

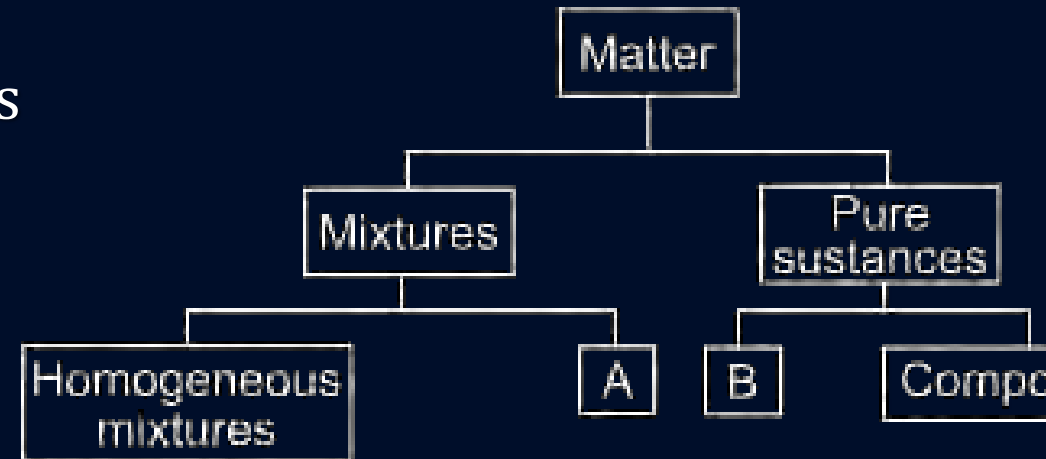
- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

Question

Matter can be classified as mixtures or pure substances. These can be further sub-divided as shown.

The missing terms A and B respectively are

- A** A → microscopic level, B → heterogeneous mixtures
- B** A → heterogeneous mixtures, B → elements
- C** A → homogeneous mixtures, B → elements
- D** A → elements, B → heterogeneous mixtures



Question

Which of the following is not a mixture?

- A** Gasoline
- B** Distilled alcohol
- C** LPG
- D** Iodised table salt

Question

Which of the following statements is/are true?

- A** An element of a substance contains only one kind of atoms.
- B** A compound can be decomposed into its components.
- C** All homogeneous mixtures are called as solutions.
- D** All of these

Question

Assertion: Components of a homogeneous mixture cannot be separated by using physical methods.

Reason: Composition of homogeneous mixture is uniform throughout as the components react to form a single compound.

- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

Question

Assertion: Elements and compounds are the examples of pure substances.

Reason: The properties of a compound are different from those of its constituent elements.

- A** If both assertion and reason are true and reason is the correct explanation of assertion.
- B** If both assertion and reason are true but reason is not the correct explanation of assertion.
- C** If assertion is true but reason is false.
- D** If both assertion and reason are false.

Question

Statement-I: Gold, silver and bromine are the pure substances.

Statement-II: The metals and non- metals are the type of elements based on the variation in property.

- A** Both statement I & II are correct
- B** Both statement I & II are incorrect
- C** Statement I is correct while statement II is incorrect
- D** Statement I is incorrect while statement II is correct

Question

Assertion (A): A gas can easily be compressed by applying pressure.

Reason (R): Since the inter-particle spaces between gases are very large, they can decrease by applying pressure.

- A** Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- B** Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- C** Assertion (A) is true and Reason (R) is false.
- D** Assertion (A) is false and Reason (R) is true.

Question

What type of a mixture is blood?

- A** Homogeneous Mixture
- B** Heterogeneous Mixture
- C** Both A and B
- D** None of these

Question

Statement-I: Many elements have non integral atomic masses.

Statement-II: Most elements have existence of isotopes which have different masses.

- A** Both statement I and II are correct
- B** Both statement I and II are incorrect
- C** Statement I is correct while statement II is incorrect
- D** Statement I is incorrect while statement II is correct

Question

Assertion (A): One atomic mass unit is defined as one twelfth of the mass of one carbon-12 atom.

Reason (R): Carbon-12 isotope is the most abundant isotope of carbon and has been chosen as standard.

- A** Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- B** Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).
- C** Assertion (A) is true and Reason (R) is false.
- D** Assertion (A) is false and Reason (R) is true.

Question

According to the Dalton's atomic theory:

- A** Electrons, proton and neutron are found in an atom
- B** Atom is indivisible
- C** Atom of various element has same properties
- D** Atoms are destroyed in the chemical reaction

Question

The molecular weight of SO_4^{2-} is:

A 192 u

B 48 u

C 96 u

D 24 u

Question

Mass of one atom of an element is 8×10^{-24} g. This is equal to

- A** 4.8 u
- B** 48 u
- C** 0.048 u
- D** 0.48 u

Question

Number of atoms of He in 100 u of He (atomic weight of He is 4) are

- A** 25
- B** 100
- C** 50
- D** $100 \times 6 \times 10^{-23}$

Question

1u is equal to

- A** $1.66 \times 10^{-24} \text{ g}$
- B** $1.66 \times 10^{-27} \text{ g}$
- C** $\frac{1}{N_A} \text{ g}$
- D** All of these

Question

The number of particles present in 1 mol of nitrogen atom are:

- A** 6.022×10^{25}
- B** 6.022×10^{24}
- C** 6.022×10^{23}
- D** 6.022×10^{22}

Question

The molar mass of ferrous sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) is:

- A** 152 gm
- B** 278 gm
- C** 137 gm
- D** None of these

Question

Choose the wrong statement.

- A** 1 mole means 6.023×10^{23} particles
- B** Molar mass is mass of one molecule
- C** Molar mass is mass of one mole of a substance
- D** Molar mass is molecular mass expressed in grams

Question

What is the mass of 0.5 moles of O_3 molecules?

A 16 gm

B 20 gm

C 40 gm

D 24 gm

Question

Two elements A and B forms 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.

Question

5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A (M_A) and molar mass of B (M_B) in kgmol^{-1} are:

- A** $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$
- B** $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
- C** $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$
- D** $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

Question

Calculate the mass of 6.022×10^{23} molecules of calcium carbonate.

- A** 50 g
- B** 100 g
- C** 200 g
- D** None of these

Question

The weight of a molecule of the compound $\text{C}_6\text{H}_{12}\text{O}_6$, is about:

- A** 180 g
- B** 3×10^{-22} g
- C** 22×10^{-22} g
- D** 132 g

Question

What is the charge of 96 amu of S^{2-} ?

- A** 2 C
- B** 3.2×10^{-19} C
- C** 9.6×10^{-19} C
- D** 6 C

Question

Number of atoms present in 49 g of H_2SO_4 ?

- A** $N_{\text{A}}/7$
- B** $7N_{\text{A}}$
- C** $7N_{\text{A}}/2$
- D** $14N_{\text{A}}$

Question

Which has maximum number of oxygen atoms?

- A** 1 gm of O
- B** 1 gm of O_2
- C** 1 gm of O_3
- D** All have same no. of O-atoms

Question

Find total number of neutrons in 56 g of CO?

If ${}^1_6\text{C}$ and ${}^{16}_8\text{O}$ if Avogadro's No. = N_A

- A** $28 N_A$
- B** $22 N_A$
- C** $11 N_A$
- D** None of these

Question

The number of water molecules in 1 L of water is:

- A** 18
- B** 18×1000
- C** N_A
- D** $55.55 N_A$

Question

Find mass of 18.066×10^{23} molecules of NH_3 ? (If atomic mass of N and H are 14 and 1)

- A** 51 g
- B** 34 g
- C** 17 g
- D** None of these

Question

Find mass of CO_2 which have same molecules as present in 68 g of NH_3 ?

- A** 88 g
- B** 176 g
- C** 44 g
- D** 132 g

Question

A sample of aluminium has a mass of 54.0 g. What is the mass of the same number of magnesium atoms? (At. wt. Al = 27, Mg = 24)

- A** 12 g
- B** 24 g
- C** 48 g
- D** 96 g

Question

A person adds 1.71 gram of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)

- A** 3.6×10^{22}
- B** 7.2×10^{21}
- C** 0.05
- D** 6.6×10^{22}

Question

5.6 litre of oxygen at STP contains

- A** 6.02×10^{23} atoms
- B** 3.01×10^{23} atoms
- C** 1.505×10^{23} atoms
- D** 0.7525×10^{23} atoms

Question

3.011×10^{22} atoms of an element weighs 1.15 gm. The atomic mass of the element is:

- A** 10 amu
- B** 2.3 amu
- C** 35.5 amu
- D** 23 amu

Question

11.2 L of $O_3(g)$ contains how many numbers of molecules?

- A** N_A molecules
- B** $N_{A/2}$ molecules
- C** $2N_A$ molecules
- D** $3N_A$ molecules

Question

The volume occupied by 4.4 g of CO_2 at STP is:

- A** 22.4 L
- B** 2.24 L
- C** 0.224 L
- D** 0.1 L

Question

2 g of O_2 at NTP occupies the volume:

- A** 1.4 L
- B** 2.8 L
- C** 11.4 L
- D** 3.2 L

Question

11.35 L of a gas at STP weighs 14 g. The gas could be:

- A** N_2O
- B** N_2
- C** CO
- D** Both (B) and (C)

Question

The maximum number of molecules is present in

- A** 15 L of H_2 gas at STP
- B** 5 L of N_2 gas at STP
- C** 1.5 g of H_2 gas
- D** 5 g of O_2 gas

Question

The number of molecules and moles in 2.8375 litres of O_2 at STP are respectively

- A** 7.527×10^{22} and 0.250 mol
- B** 1.505×10^{23} and 0.250 mol
- C** 7.527×10^{23} and 0.125 mol
- D** 7.527×10^{22} and 0.125 mol

Question

Total number of neutrons present in 4 g of heavy water (D_2O) is: (Where N_A represents Avogadro's number)

- A** $2.4 N_A$
- B** $4 N_A$
- C** $1.2 N_A$
- D** $2 N_A$

Question

In which of the following options, the molecules are correctly matched with their atomicity?

(1) P_4	(p) 7
(2) HNO_3	(q) 6
(3) N_2H_4	(r) 5
(4) H_2SO_4	(s) 4

A (1-s), (2-r), (3-p), (4-q)

B (1-q), (2-s), (3-p), (4-r)

C (1-q), (2-s), (3-r), (4-p)

D (1-s), (2-r), (3-q), (4-p)

Question

In which of the following pairs, do 1 g of each have an equal number of molecules?

- A** N_2O and CO
- B** N_2 and CO
- C** N_2O and CO_2
- D** Both (B) and (C)

Question

**The mass of half mole of electrons is about:
(Given : Mass of electron = 9.1×10^{-28} g)**

- A** 0.546 mg
- B** 0.273 mg
- C** 1.092 mg
- D** 4.55 mg

Question

The charge on 1 gram of Al^{3+} ions is: (e = magnitude of electronic charge)

A $\frac{1}{27} N_A e$ coulomb

B $\frac{1}{3} N_A e$ coulomb

C $\frac{1}{9} N_A e$ coulomb

D $N_A e$ coulomb

Question

The atomic weight of an element is 'a'. If this element occurs in nature as a triatomic gas, then the correct formula for the number of moles of gas in its 'w' g is:

A $\frac{3w}{a}$

B $\frac{w}{3a}$

C $3wa$

D $\frac{a}{3w}$

Question

A gaseous mixture contains oxygen and sulphur dioxide in the ratio of 1 : 4 by mass. Therefore, the ratio of their respective number of molecules is:

A 1 : 2

B 1 : 4

C 2 : 1

D 4 : 1

Question

Sulphur exist in different allotropic forms like S_2 , S_6 and S_8 etc. If equal moles of these three forms are taken in separate containers, then the ratio of number of atoms present in them respectively is:

- A** 1 : 3 : 4
- B** 1 : 1 : 1
- C** 12 : 4 : 3
- D** 4 : 3 : 1

Question

The number of F^- ions in 2.1 g AlF_3 is. ($N_A = 6 \times 10^{23}$)

- A** 1.5×10^{22}
- B** 6×10^{22}
- C** 3×10^{22}
- D** 4.5×10^{22}

Question

An element is found in nature in two isotopic forms with mass numbers $(A - 1)$ and $(A + 3)$. If the average atomic mass of the element is found to be A , then the relative abundance of the heavier isotope in the nature will be:

- A** 66.6%
- B** 75%
- C** 25%
- D** 33.3%

Question

A balanced chemical equation is in accordance with-

- A** Avogadro's law
- B** Law of constant proportions
- C** Law of conservation of mass
- D** Law of gaseous volumes

Question

10 g CaCO_3 on heating gives 5.6 g CaO and ... g CO_2 .

A 4.4

B 5.6

C 6.5

D 4.2

Question

The law stating that a given compound always contains exactly the same proportion of elements by weight was given by.

- A** Lavoisier
- B** Proust
- C** Dalton
- D** Avogadro

Question

Percentage of copper and oxygen in sample of CuO obtained by different methods were found to be same. This proves the law of-

- A** Constant proportion
- B** Multiple proportion
- C** Reciprocal proportion
- D** None of these

Question

Two gaseous samples were analyzed. One contained 1.2 g of carbon and 3.2 g of oxygen. The other contained 27.3% carbon and 72.7% oxygen. The experimental data are in accordance with

- A** Law of conservation of mass
- B** Law of definite proportions
- C** Law of reciprocal proportion
- D** Law of multiple proportion

Question

A sample of calcium carbonate (CaCO_3) has the following percentage composition:

Ca = 40%, C = 12%, O = 48%.

If the law of constant proportions is true, then the weight of calcium in 4 g of a sample of calcium carbonate obtained from another source will be:

- A** 0.016 g
- B** 0.16 g
- C** 1.6 g
- D** 16 g

Question

The percentage of hydrogen in water and hydrogen peroxide is 11.1 and 5.9% respectively. These figures illustrate-

- A** Law of multiple proportions
- B** Law of conservation of mass
- C** Law of reciprocal proportions
- D** Law of combining volumes

Question

Nitrogen forms five stable oxides with oxygen of the formula, N_2O , NO , N_2O_3 , N_2O_4 , N_2O_5 . The formation of these oxides explains fully the-

- A** Law of definite proportions
- B** Law of partial pressures
- C** Law of multiple proportions
- D** Law of reciprocal proportions

Question

The law of multiple proportions is illustrated by-

- A** Carbon monoxide and carbon dioxide
- B** Potassium bromide and potassium chloride
- C** Water and heavy water
- D** Calcium hydroxide and barium hydroxide

Question

Two elements X (of mass 16) and Y (of mass 14) combine to form compounds A, B and C. The ratio of different masses of Y which combine with a fixed mass of X in A, B and C is 1 : 3 : 5, if 32 parts by mass of X combines with 84 parts by mass of Y in B, then in C, 16 parts by mass of X will combine with;

- A** 14 parts by mass of Y
- B** 42 parts by mass of Y
- C** 70 parts by mass of Y
- D** 84 parts by mass of Y

Question

Which one of the following pairs of compounds illustrates the law of multiple proportion?

- A** H_2O , Na_2O
- B** Na_2O , BaO
- C** MgO , Na_2O
- D** SnCl_2 , SnCl_4

Question

Carbon + oxygen → Carbon dioxide

12g 32g 44g

Carbon + Oxygen → Carbon monoxide

12g 16g 28g

This data illustrates which of the following law

- A** Law of multiple proportion
- B** Law of reciprocal proportion
- C** Law of conservation of mass
- D** Both (A) and (C)

Question

1L of N_2 combines with 3L of H_2 to form 2L of NH_3 under the same conditions. This illustrates the-

- A** Law of constant composition
- B** Law of multiple proportions
- C** Law of reciprocal proportions
- D** Gay Lussac's law of gaseous volumes

Question

Gay Lussac's law is not valid in the chemical reaction:



Question

Equal volume of gases contains equal number of molecules at same temp and pressure, this law was given by

- A** Berzillius
- B** Gay - Lussac
- C** Dalton
- D** Avogadro

Question

What must be held constant when applying Avogadro's law?

- A** pressure and temperature
- B** volume and temperature
- C** moles and temperature
- D** pressure and volume

Question

Which one of the following represents Avogadro's hypothesis?

- A** Gases react together in volumes which bear a simple ratio to one another
- B** Equal volumes of all gases under same conditions of temperature and pressure contain equal number of molecules
- C** Equal volumes of all gases under same conditions of temperature and pressure contain equal number of atoms
- D** The rates of diffusion of gases are inversely proportional to the square root of their densities

Question

6g of carbon combines with 32g of sulphur to form CS_2 . 12g of C also combine with 32g of oxygen to form carbon dioxide. 10g of sulphur combines with 20g of oxygen to form sulphur dioxide. Which law is illustrated by the above example-

- A** Law of multiple proportions
- B** Law of constant composition
- C** Law of reciprocal proportions
- D** Gay Lussac's law

Question

Hydrogen combines with oxygen to form H_2O in which 16 g of oxygen combine with 2 g of hydrogen. Hydrogen also combines with carbon to form CH_4 in which 2 g of hydrogen combine with 6 g of carbon. If carbon and oxygen combine together then they will do in the ratio of

A 6 : 16 or 12 : 32

B 1 : 2

C 6 : 18

D 12 : 24

Question

One of the following combinations which illustrates the law of reciprocal proportions is:

- A** $\text{N}_2\text{O}_3, \text{N}_2\text{O}_4, \text{N}_2\text{O}_5$
- B** $\text{NaCl}, \text{NaBr}, \text{NaI}$
- C** $\text{CS}_2, \text{CO}_2, \text{SO}_2$
- D** $\text{PH}_3, \text{P}_2\text{O}_3, \text{P}_2\text{O}_5$

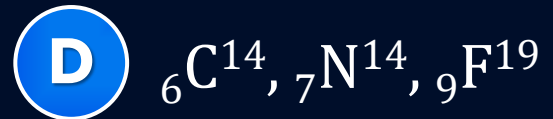
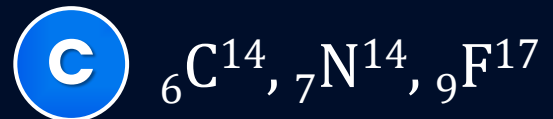
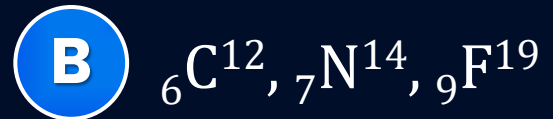
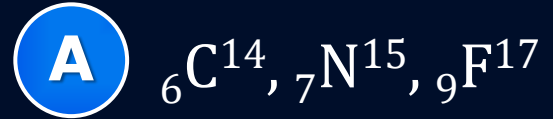
Question

Dalton's Atomic theory could not explain which of the following?

- A** Law of multiple proportion
- B** Law of gaseous volume
- C** Law of conservation of mass
- D** Law of constant proportion

Question

The triad of nuclei which is isotonic will be



Question

11.2 litre of a gas at STP weighs 14 g. The gas could not be:

- A** N_2
- B** CO
- C** C_2H_4
- D** N_2O

Question

"A + 2" is isoelectronic with N_2O and has $(Z + 1)$ neutron (Z is atomic number of A) then mass number of A is:

- A** 45
- B** 41
- C** 49
- D** 43

Question

The atomic weights of two elements A and B are 40 and 80 respectively. If x g of A contains y atoms, how many atoms are present in $2 \times x$ g of B?

A $\frac{y}{2}$

B $\frac{y}{4}$

C y

D $2y$

Question

X and Y are two elements which form X_2Y_3 and X_3Y_4 . If 0.20 mol of X_2Y_3 weighs 32.0 g and 0.4 mol of X_3Y_4 weighs 92.8 g, the atomic weights of X and Y are respectively.

A 16.0 and 56.0

B 8.0 and 28.0

C 56.0 and 16.0

D 28.0 and 8.0

Question

X g Urea (NH_2CONH_2) contains y total number of atoms. 4X g Acetic acid (CH_3COOH) contains py total number of atoms. Find the value of p.

- A** 8
- B** 4
- C** 2
- D** 6

Question

The pairs which illustrate the law of multiple proportions is.

- A** PH_3 , HCl
- B** CuCl_2 , CuSO_4
- C** PbO , PbO_2
- D** H_2S , SO_2

Question

Which of the following reactions is not correct according to the law of conservation of mass?

- A** $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$
- B** $\text{C}_3\text{H}_8(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- C** $\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$
- D** $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$

Question

A bivalent metal has an atomic mass of 64u. The molecular mass of the metal nitrate is.

- A** 182
- B** 168
- C** 192
- D** 188

Question

If N_A is Avogadro's number then number of electrons in 4.2 g of N^{-3} :

A $2.1 N_A$

B $3 N_A$

C $6 N_A$

D N_A

Question

An oxide (Fe_2O_n) of iron contains 70% iron by mass. Calculate value of n .

- A** 1
- B** 2
- C** 3
- D** 4

Question

The percentage of oxygen in NaOH is:

- A** 40
- B** 8
- C** 60
- D** 10

Question

The mass per cent of different elements present in sodium sulphate, (Na_2SO_4) respectively are

- A** 32.37; 45.06 and 22.57
- B** 22.57; 32.37 and 45.06
- C** 45.06; 32.37 and 40.06
- D** 32.37; 22.57 and 45.06

Question

What is the mass per cent of carbon in carbon dioxide?

- A** 0.034%
- B** 27.27%
- C** 3.4%
- D** 28.7%

Question

$\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$ has 50% H_2O by mass. Hence, x is:

A 4

B 5

C 6

D 8

Question

The hydrated salt $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ undergoes 63% loss in mass on heating and becomes anhydrous. The value of x is:

- A** 10
- B** 12
- C** 8
- D** 18

Question

An organic compound made of C, H and N, contains 20% nitrogen. Its minimum molecular mass is:

- A** 50
- B** 70
- C** 65
- D** 100

Question

Caffeine has a molecular weight of 194. It contains 28.9% by mass of nitrogen.
Number of atoms of nitrogen in one molecule of it:

- A** 2
- B** 3
- C** 4
- D** 5

Question

Statement-I: The percentage of nitrogen in urea is 46.6%.

Statement-II: Urea is an ionic compound.

- A** Both Statement-I and Statement-II are correct.
- B** Both Statement-I and Statement-II are incorrect.
- C** Statement-I is correct and Statement-II is incorrect.
- D** Statement-I is incorrect and Statement-II is correct.

Question

A 100 g sample of Haemoglobin on analysis was found to contain 0.34% Fe by mass. If each haemoglobin molecule has four Fe^{2+} ions, the molecular mass of haemoglobin is ($\text{Fe} = 56\text{amu}$):

- A** 77099.9 g
- B** 12735 g
- C** 96359.9 g
- D** 65882 g

Question

**Calculate the empirical formula if a compound has C, H & N in the ratio 9 : 1 : 3.5
Also calculate its Molecular Formula if Molecular mass is 108.**

Question

The empirical formula and molecular mass of a compound are CH_2O and 180 g respectively. What will be the molecular formula of the compound?



Question

A compound made of two elements A and B are found to contain 25% A (at mass 12.5) and 75% B (at mass 37.5). The simplest formula of the compound is:

- A** AB
- B** AB₂
- C** AB₃
- D** A₃B

Question

A certain compound contains magnesium, carbon and Nitrogen in the mass ratio 12 : 12 : 14. The formula of the compound is:

- A** MgCN
- B** Mg_2CN
- C** MgCN_2
- D** $\text{Mg}(\text{CN})_2$

Question

A hydrocarbon contains 85.7% of Carbon and 14.3% of Hydrogen. If 42 mg of the compound contains 3.01×10^{20} molecules, the molecular formula of the compound will be:

- A** C_2H_4
- B** C_3H_6
- C** C_6H_{12}
- D** $\text{C}_{12}\text{H}_{24}$

Question

0.30 g of an organic compound containing C, H and O on combustion yields 0.44 g CO_2 and 0.18 g H_2O . If one mole of compound weighs 60 g, then molecular formula of the compound is:

- A** $\text{C}_3\text{H}_8\text{O}$
- B** $\text{C}_2\text{H}_4\text{O}_2$
- C** CH_2O
- D** $\text{C}_4\text{H}_6\text{O}$

Question

45.4 L of dinitrogen reacted with 22.7 L of dioxygen and 45.4 L of nitrous oxide was formed. The reaction is given below.



Which law is being obeyed in this experiment?

- A** Law of conservation of mass
- B** Law of definite proportion
- C** Law of multiple proportion
- D** Gay Lussac's law of gaseous volumes

Question

Which of the following statements about a compound is/are incorrect?

- I. A molecule of a compound has atoms of different elements.**
- II. A compound cannot be separated into its constituent elements by physical methods of separation.**
- III. A compound retains the physical properties of its constituent elements.**
- IV. The ratio of atoms of different elements in a compound is fixed.**

- A** Only III
- B** Only II
- C** I and IV
- D** III and IV

Question

Among the following statements, that which was not proposed by Dalton was

- A** chemical reactions involve reorganisation of atoms. These are neither created nor destroyed in a chemical reaction.
- B** when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T and P.
- C** all the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.
- D** matter consists of indivisible atoms.

Question

Which of these is a necessary condition for law of definite proportions to hold true?

- A** The compound under observation must not sublime
- B** The compound under observation must be pure
- C** The compound should not crack on heating
- D** The compound should not be radioactive

Question

Which of these is not a violation of conservation of mass?

- A** When petrol is combusted in a vessel, finally the vessel is empty
- B** When a piece of camphor is left standing in a dish, after a long time the dish is empty
- C** When a bottle of chloroform is left open, after a short time the bottle has nothing in it.
- D** None of the above choices is a violation of the law.

Question

Which of these statements is incorrect regarding the laws of chemical combinations?

- A** Combination of elements in ratio by atoms is same as their combination ratio by mass
- B** Elements may not combine with each other also. For instance, sodium and potassium don't combine to form compounds.
- C** Elements can have multiple valencies
- D** Fixed amount of an element should combine with different amounts of another element for law of multiple proportions.

Question

Which of the following proves that law of definite proportions is not universally true?

- A** The discovery of multiple compounds formed between the same two elements
- B** The discovery of non-stoichiometric compounds
- C** The discovery of networked solids
- D** The discovery of salt-like carbides

Question

What mass of barium chloride would be decomposed by 9.8 g of sulphuric acid if 12 g of barium sulphate and 2.75 g of hydrogen chloride were produced in a reaction assuming that the law of conservation of mass is true?

- A** 5.6 g
- B** 7.8 g
- C** 3.1 g
- D** 4.95 g

Question

Myoglobin stores oxygen for metabolic processes in muscle. Chemical analysis shows that it contains 0.34% Fe by mass. Minimum molecular mass of myoglobin is:

- A** 16470 gm
- B** 16176 gm
- C** 17500 gm
- D** 1647 gm

Question

The pair of species having same percentage (mass) of carbon is:

- A** CH_3COOH and $\text{C}_6\text{H}_{12}\text{O}_6$
- B** CH_3COOH and $\text{C}_2\text{H}_5\text{OH}$
- C** HCOOCH_3 and $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- D** $\text{C}_6\text{H}_{12}\text{O}_6$ and $\text{C}_{12}\text{H}_{22}\text{O}_{11}$

Question

0.30 g of an organic compound containing C, H and O on combustion yields 0.44 g CO_2 and 0.18 g H_2O . If one mole of compound weighs 60 g, then molecular formula of the compound is:

- A** $\text{C}_3\text{H}_8\text{O}$
- B** $\text{C}_2\text{H}_4\text{O}_2$
- C** CH_2O
- D** $\text{C}_4\text{H}_6\text{O}$

Question

An oxide of iodine (atomic mass = 127) contains 25.4 g of iodine and 8 g of oxygen. Its formula would be:



Question

The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is:



Question

An oxide of sulphur contains 50% of sulphur in it. Its empirical formula is:



Question

The carbonate of a metal is isomorphous with MgCO_3 and contains 10.34 percent of carbon. Atomic mass of metal will be:

- A** 56
- B** 98
- C** 40
- D** 23

Question

1 atm = _____ bar:

- A** 760
- B** 76
- C** 1.013
- D** 1000

Question

A scientific instrument used to measure air pressure is called:

- A** Hygrometer
- B** Sphygmomanometer
- C** Lactometer
- D** Barometer

Question

The S.I. unit of pressure is:

- A** Atmosphere
- B** Bar
- C** Pascal
- D** Torr

Question

Total number of moles of oxygen atoms in 3 litre $\text{O}_3(\text{g})$ at 27°C and 8.21 atm are:

- A** 3
- B** 1
- C** 2
- D** None of these

Question

If two moles of a ideal gas at 546 K occupy volume 44.8 L, then pressure must be

- A** 2 atm
- B** 3 atm
- C** 4 atm
- D** 1 atm

Question

For an ideal gas, number of mol per litre in terms of its pressure p , temperature T and gas constant R is.

A pT/R

B pRT

C p/RT

D RT/p

Question

5.6 L of a gas at S.T.P. weights equal to 8 g. The vapour density of gas is:

- A** 32
- B** 16
- C** 8
- D** 40

Question

Vapour density of a gas is 16. The ratio of specific heat at constant pressure to specific heat at constant volume is 1.4, then its atomic weight is:

- A** 8
- B** 16
- C** 24
- D** 32

Question

The vapour density of gas A is three times that of gas B. If the molar mass of A is M , the molar mass of B is:

- A** $3M$
- B** $\sqrt{3} M$
- C** $M/3$
- D** $M/\sqrt{3}$

Question

Density of a gas STP is 1.43 g L^{-1} . What is its molecular weight?

- A** 16
- B** 32
- C** 48
- D** 64

Question

Vapour density of a diatomic elementary gas is 16. The atomic mass of gas is _____

- A** 16
- B** 17
- C** 19
- D** 10

Question

The vapour density of a gas A is twice that of a gas B. If the molecular weight of B is M , the molecular weight of A will be:

- ☐ A M
- ☐ B $2M$
- ☐ C $3M$
- ☐ D $M/2$

Question

A certain mass of a gas occupies a volume of 2 L at STP. To what temperature the gas must be heated to double its volume, keeping the pressure constant?

A 100 K

B 273 K

C 273°C

D 546°C

Question

Correct gas equation is

A $\frac{p_1 T_1}{V_1} = \frac{p_2 T_2}{V_2}$

B $\frac{V_1 T_2}{p_1} = \frac{V_2 T_2}{p_2}$

C $\frac{p_1 T_1}{p_2 V_2} = \frac{T_1}{T_2}$

D $\frac{V_1 V_2}{T_1 T_2} = p_1 p_2$

Question

If the absolute temperature of a gas is doubled and the pressure is reduced to one half, the volume of the gas will

- A** Remain unchanged
- B** Be doubled
- C** Increase four-fold
- D** Be halved

Question

The density of a gaseous element is 5 times that of oxygen under similar conditions. If the molecule of the element is triatomic, what will be its atomic mass?

- A** 32
- B** 160
- C** 53.3
- D** 80

Question

The molecular weight of O_2 and SO_2 are 32 and 64 respectively. If one litre of O_2 at $15^\circ C$ and 750 mm pressure contains 'N' molecules, the number of molecules in two litres of SO_2 under the same conditions of temperature and pressure will be:

- A** $N/2$
- B** N
- C** $2N$
- D** $4N$

Question

The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is.

- A** 40
- B** 10
- C** 20
- D** 30

Question

Calculate the mass of hydrogen formed when 27 g of aluminum reacts with excess hydrochloric acid according to the balanced equation below.



A 1.5 g

B 2.0 g

C 3.0 g

D 6.0 g

Question

Equal mass of oxygen, hydrogen and methane are taken in a container in identical conditions. What is the ratio of their moles:

A 1 : 16 : 1

B 1 : 16 : 2

C 8 : 1 : 8

D 16 : 1 : 8

Question

A gas is found to have the formula $(\text{CO})_x$. Its VD is 70 the value of x must be:

A 7

B 4

C 5

D 6

Question

How many grams of H_2 are required to consume 2 mol of CO ?



- A** 2 g
- B** 4 g
- C** 8 g
- D** 16 g

Question

The moles of methane required to produce 81 g of water after complete combustion is _____ $\times 10^{-2}$ mol. (nearest integer)

Question

The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is_____.

Question

How many grams of SO_3 are produced from 1 mol of S_8 ?

- A** 1280.0
- B** 640.0
- C** 960.0
- D** 320.0

Question

If $1/2$ moles of oxygen combine with aluminium to form Al_2O_3 then weight of Aluminium metal used in the reaction is (Al = 27)

- A** 27 g
- B** 18 g
- C** 54 g
- D** 40.5 g

Question

When a certain amount of octane, C_8H_{18} , is burnt completely, 7.04 g CO_2 is formed. What is the mass of H_2O formed simultaneously?

- A** 1.62 g
- B** 6.48 g
- C** 3.24 g
- D** 2.28 g

Question

What volume at STP of CO is required to reduce one mole of Fe_2O_3 in the following reaction.



- A** 11200 cm^3
- B** 22400 cm^3
- C** 67200 cm^3
- D** 33600 cm^3

Question

The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is.

(Atomic mass of Ca = 40)



- A** 1.76 g
- B** 2.64 g
- C** 1.32 g
- D** 1.12 g

Question

A sample of calcium carbonate is 80% pure. 25g of this sample is treated with excess of HCl. How much volume of CO_2 will be obtained at NTP.

A 4.48 litre

B 5.6 litre

C 11.2 litre

D 2.24 litre

Question

The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is ____kg. (Nearest integer)

Question

20.0 g of a magnesium carbonate sample decompose on heating to give carbon dioxide and 8.0 g magnesium oxide. What be the percentage purity of magnesium carbonate in the sample? [2015]

- A** 60
- B** 84
- C** 75
- D** 96

Question

4g CaCO_3 on reacting with excess HCl produces 0.88 g CO_2 according to reaction:



The percentage purity of CaCO_3 :

- A** 50 %
- B** 60 %
- C** 70 %
- D** 80 %

Question

10g of MnO_2 on reaction with HCl forms 2.24 L of $\text{Cl}_2(\text{g})$ at NTP, the percentage impurity of MnO_2 is.



- A** 87%
- B** 25%
- C** 33.3%
- D** 13%

Question

If 300 g of CaCO_3 on reaction with excess of H_2SO_4 gives 272 g of CaSO_4 . Find the % age yield of product?

A 66.66%

B 33.33%

C 50%

D 25%

Question

Find the number of moles of C formed when 20 mole of A react with excess of B as per the reaction:

$2A + B \rightarrow 3C$, If the percentage yield is 80%.

A 30

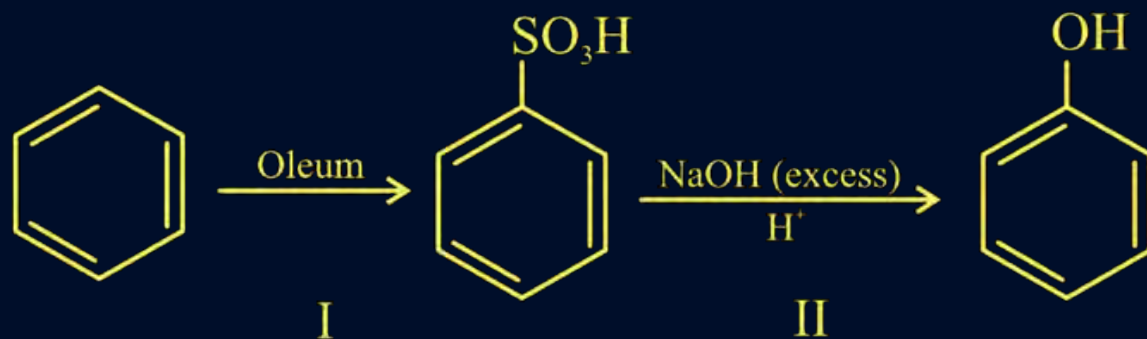
B 24

C 20

D 28

Question

In the following reaction



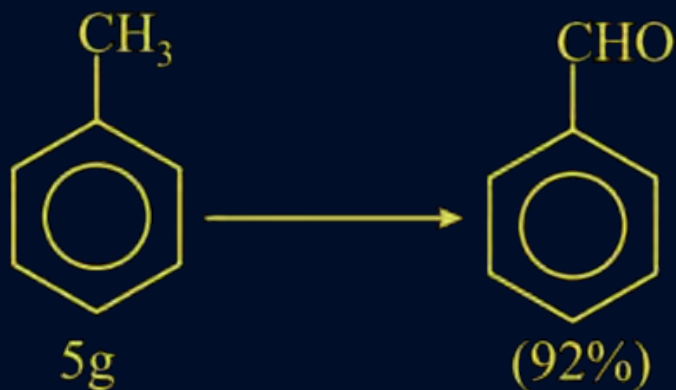
The % yield for reaction I is 60% and that of reaction II is 50%. The overall yield of the complete reaction is _____ % [Nearest integer]

Question

Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) (Nearest Integer)

Question

In the below reaction, 5 g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is $\text{_____} \times 10^{-2}$ g. (Nearest integer)



Question

When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?



Question

When a hydrocarbon A undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water. What is the molecular formula of A?

- A** C_9H_8
- B** C_{11}H_4
- C** C_5H_8
- D** C_{11}H_8

Question

40 ml of a hydrocarbon undergoes combustion in 260 ml of oxygen and gives 160 ml of carbon dioxide. If all gases are measured under similar conditions of temperature and pressure, the formula of hydrocarbon is:

- A** C_3H_8
- B** C_4H_8
- C** C_6H_{14}
- D** C_4H_{10}

Question

Limiting reagent in a chemical reaction is that reactant which

- A** Left some amount unreacted after the completion of reaction
- B** Reacts completely in the reaction
- C** Does not react in the reaction
- D** All of these

Question

When 5 mole N_2 react with 10 mole H_2 in Haber's process, the mole of NH_3 formed will be.

- A** 10 mole
- B** $10/3$ mole
- C** $20/3$ mole
- D** 5 mole

Question

When 22.4 L H_2 (g) is mixed with 11.2 L Cl_2 (g) at STP the moles of HCl formed are equal to

- A** 1 mole of HCl (g)
- B** 2 mole of HCl (g)
- C** 0.5 mole of HCl (g)
- D** 1.5 mole of HCl (g)

Question

What is the maximum amount of nitrogen dioxide that can be produced by mixing 4.2 gm of NO(g) and 3.2 gm of $\text{O}_2\text{(g)}$?

- A** 4.60 gm
- B** 2.30 gm
- C** 3.22 gm
- D** 6.44 gm

Question

According to the following reaction the minimum quantity in gm of H_2S needed to precipitate 63.5 gm of Cu^{2+} ions will be nearly



- A** 63.5 g
- B** 31.75 gm
- C** 34 gm
- D** 20 gm

Question

12 moles of each A and B are allowed to react as given: $3A + 2B \rightarrow C + \frac{1}{2} D$.
If 60g of D is produced then calculate the atomic mass of D.

A 30

B 45

C 60

D 15

Question

1.0 g magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and how much?

(At. wt. Mg = 24, O = 16)

- A** Mg, 0.16 g
- B** O_2 , 0.16 g
- C** Mg, 0.44 g
- D** O_2 , 0.28 g

Question

In a chemical reaction, $A + B \rightarrow AB$, if B is acting as a limiting reagent, then choose the correct option.

A

B

A 50 atoms

100 atoms

B 100 atoms

200 atoms

C 50 atoms

30 atoms

D 50 atoms

200 atoms

Question

If ' x ' dm³ of N₂ reacts with ' x ' dm³ of H₂ to form NH₃ under suitable conditions, then

- A** H₂ is limiting reagent
- B** N₂ is limiting reagent
- C** Both reactants are limiting reagents
- D** NH₃ is limiting reagent

Question

A mixture of 1.0 mole of Al and 3.0 mole of Cl_2 are allowed to react as:



(1) Which is limiting reagent?

(2) How many moles of AlCl_3 are formed

(3) Moles of excess reagent left unreacted is

A (1) Al, (2) 1.0 (3) 1.5

B (1) Cl_2 , (2) 2.0 (3) 2.0

C (1) Al, (2) 0.5 (3) 1.5

D (1) Cl_2 , (2) 1.0 (3) 1.5

Question

In Haber's process 30 litre of dihydrogen and 30 L of dinitrogen were taken for reaction which yielded only 50% of expected product. What will be the composition of gaseous mixture under the aforesaid condition in the end?

- A** $\text{N}_2(\text{g}) = 20 \text{ L}; \text{H}_2(\text{g}) = 0; \text{NH}_3(\text{g}) = 20 \text{ L}$
- B** $\text{N}_2(\text{g}) = 25 \text{ L}; \text{H}_2(\text{g}) = 30 \text{ L}; \text{NH}_3(\text{g}) = 10 \text{ L}$
- C** $\text{N}_2(\text{g}) = 25 \text{ L}; \text{H}_2(\text{g}) = 15 \text{ L}; \text{NH}_3(\text{g}) = 10 \text{ L}$
- D** $\text{N}_2(\text{g}) = 10 \text{ L}; \text{H}_2(\text{g}) = 25 \text{ L}; \text{NH}_3(\text{g}) = 15 \text{ L}$

Question

4 g of hydrogen is ignited with 4 g of oxygen, the amount of water formed is?

- A** 2.5 g
- B** 0.5 g
- C** 4.5 g
- D** 8 g

Question

Equal volume of N_2 and H_2 react to form ammonia under suitable condition then the limiting reagent is.

- A** H_2
- B** NH_3
- C** N_2
- D** No one reactant is limiting reagent

Question

For the reaction:



If 15 moles of A, 26 moles of B & 30.5 moles of C are taken initially then limiting reactant is.

- A** A
- B** B
- C** C
- D** None of these

Question

Zinc and hydrochloric acid react according to the reaction.



If 0.30 mole of Zn are added to hydrochloric acid containing 0.52 mole HCl, how many moles of H_2 are produced?

A 0.26

B 1.04

C 0.52

D 0.13

Question

8 g H_2 and 32 g O_2 is allowed to react to form water then which of the following statement is correct?

- A** O_2 is limiting reagent
- B** O_2 is reagent in excess
- C** H_2 is limiting reagent
- D** 40 g water is formed

Question

0.5 mole of H_2SO_4 is mixed with 0.2 mole of $\text{Ca}(\text{OH})_2$. The maximum number of moles of CaSO_4 formed is.

A 0.2

B 0.5

C 0.4

D 1.5

Question

10 mL of gaseous hydrocarbon on combustion gives 40 mL of $\text{CO}_2(\text{g})$ and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is _____.

Question

120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are:

- A** 25 and 75
- B** 40 and 60
- C** 60 and 40
- D** 75 and 25

Question

The weight of lime obtained by heating 200 kg of 95% pure lime stone is.

- A** 98.4 kg
- B** 106.4 kg
- C** 112.8 kg
- D** 122.6 kg

Question

10g of MnO_2 on reaction with HCl form 2.24L of $\text{Cl}_2(\text{g})$ at NTP. The percentage purity of MnO_2 is $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$.

A 87%

B 80%

C 27%

D 15%

Question

56.0 L of nitrogen gas is mixed with excess of hydrogen gas and it is found that 20 L of ammonia gas is produced. The volume of unused nitrogen gas is found to be _____L.

Question

5 moles of A, 6 moles of Z are mixed with sufficient amount of C to finally produce F. Then find the maximum moles of F which can be produced. Assuming that the product formed can also be reused. Reactions are:



A 3 moles

B 4.5 moles

C 5 moles

D 6 moles

Question

Calculate the volume by volume percentage of a solution of 15 ml of alcohol in 60 ml of water.

- A** 20
- B** 25
- C** 30
- D** 50

Question

Determine the volume/volume percent solution made by containing 25 mL of ethanol with enough water to produce 200 mL of the solution:

- A** 12.5
- B** 20
- C** 40
- D** 25

Question

Rashida dissolved 40 g of sugar in 600 mL of sugar solution. Calculate the mass by volume percentage.

A 66.6

B 70

C 6.66

D 50

Question

Given, 5×10^{-3} kg of urea of dissolved in 2×10^{-3} kg of water. Calculate the percent by mass of urea.

A 90%

B 71.42%

C 70%

D 80%

Question

What is conc. in ppm if 0.025 g of KCl is dissolved in 100 grams of water?

- A** 25 ppm
- B** 250 ppm
- C** 2.5 ppm
- D** 0.25 ppm

Question

A 500 gm toothpaste sample has 0.2 g fluoride concentration. The conc. of fluoride ions in terms of ppm.

- A** 25
- B** 250
- C** 400
- D** 40

Question

Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____ (Nearest Integer)
(Given: Atomic mass of Mg is 24 g mol^{-1} ; $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)

Question

Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is_____

Atomic weight: Fe = 55.85; S = 32.00; O = 16.00

Question

A solution is prepared by adding 360 g of glucose to 864 g of water. Calculate mole fraction of glucose (molar mass of glucose = 180)

A 0.02

B 0.04

C 0.5

D 10.2

Question

Which of the following is correct?

- A** The sum of mole fractions of all the components in a solution is always unity
- B** Mole fraction depends upon temperature
- C** Mole fraction is always negative
- D** Mole fraction is independent of content of solute in solution.

Question

What is the quantity of water that should be added to 16 g. methanol to make the mole fraction of methanol as 0.25?

- A** 27 g
- B** 12 g
- C** 18 g
- D** 36 g

Question

Molality is expressed in units of

- A** mol kg^{-1}
- B** mol L^{-1}
- C** $\text{mol L}^{-1} \text{ s}^{-1}$
- D** $\text{mol l g}^{-1} \text{ s}^{-1}$

Question

Calculate the molality of solution containing 3 g glucose dissolved in 30 g of water. (molar mass of glucose = 180)

- A** 0.50 m
- B** 0.56 m
- C** 0.091 m
- D** 0.05 m

Question

**3.42 g of a substance of molecular weight 342 g is present in 250 g of water.
Molality of this solution is:**

- A** 0.04 m
- B** 0.4 m
- C** 4.0 m
- D** 0.8 m

Question

Mole fraction of solvent in aqueous solution of NaOH having molality of 3 is:

- A** 0.3
- B** 0.05
- C** 0.7
- D** 0.95

Question

Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is:

- A** 13.8
- B** 15.5
- C** 14.5
- D** 16.8

Question

Which of the following methods of expressing concentration varies with temperature?

- A** Molality
- B** Weight percent
- C** Molarity
- D** Mole fraction

Question

If 1.8 gm glucose is present in 200 ml of solutions. Calculate molarity.

- A** 0.05 M
- B** 1.0 M
- C** 2.0 M
- D** 0.25 M

Question

A given solution of NaOH contains 4.00 g of NaOH per litre of solution. Calculate the molarity of this solution.

- A** 0.1
- B** 0.5
- C** 0.9
- D** 1.5

Question

6.025×10^{20} molecules of acetic acid are present in 500 ml of its solution. The concentration of solution is:

- A** 0.002 M
- B** 10.2 M
- C** 0.012 M
- D** 0.001 M

Question

What volume of a 0.8 M solution contains 100 millimoles of the solute?

- A** 100 mL
- B** 125 mL
- C** 500 mL
- D** 62.5 mL

Question

The molarity of the solution containing 2.8% mass-volume solution of KOH/(Given atomic mas K – 39) is:

- A** 0.1 M
- B** 0.5 M
- C** 0.2 M
- D** 1 M

Question

Concentration of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in normal blood is approximately 90 mg per 100 mL. What is the molarity of the glucose solution in blood?

- A 5 M
- B 0.005 M
- C 0.05 M
- D 1 M

Question

How many grams of NaOH should be added to water to prepare 250 ml solution of 2 M NaOH?

- A** 9.6×10^3
- B** 2.4×10^3
- C** 20
- D** 24

Question

Concentrated aqueous sulphuric acid is 98% H_2SO_4 (w/v) and has a density of 1.80 g mL^{-1} . Molarity of solution.

A 1 M

B 1.8 M

C 10 M

D 1.5 M

Question

The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% (H_2SO_4 molar mass = 98 g mol^{-1}) by mass will be:

- A** 1.22
- B** 1.45
- C** 1.64
- D** 1.88

Question

Aqueous solution of a solute X [molar mass = 60] has density 0.98 g/ml
If the molality of solution is 3 molal, the molarity solution is:

- A** 3 M
- B** 2.49 M
- C** 3.02 M
- D** 3.05 M

Question

What volume of water is to be added to 100 cm^3 of 0.5 M NaOH solution to make it 0.1 M solution?

- A** 200 cm^3
- B** 400 cm^3
- C** 500 cm^3
- D** 100 cm^3

Question

Density of a solution containing $x\%$ by mass of H_2SO_4 is y . The normality is;

A $\frac{xy \times 10}{98}$

B $\frac{xy \times 10}{98y} \times 2$

C $\frac{xy \times 10}{98} \times 2$

D $\frac{x \times 10}{98y}$

Question

The amount of oxalic acid (eq. wt. 63) required to prepare 500 ml of its 0.05 N solution is:

- A** 0.315 g
- B** 1.575 g
- C** 6.300 g
- D** 63.00 g

Question

Equivalent weight of N_2 in the change $\text{N}_2 \rightarrow \text{NH}_3$ is

A $\frac{28}{6}$

B 28

C $\frac{28}{2}$

D $\frac{28}{3}$

Question

When HNO_3 is converted into NH_3 , the equivalent weight of HNO_3 will be

- A** $M/2$
- B** $M/1$
- C** $M/6$
- D** $M/8$

Question

If 2.4 g of a metal displaces 0.1 g of hydrogen gas then find equivalent weight of metal.

Question

74.5 g of metallic chloride contains 35.5g of chlorine, find equivalent weight of metal.

Question

B_1 g of an element gives B_2 g of its chloride, the equivalent mass of the element is:

A $\frac{B_1}{B_2 - B_1} \times 35.5$

B $\frac{B_2}{B_2 - B_1} \times 35.5$

C $\frac{B_2 - B_1}{B_1} \times 35.5$

D $\frac{B_2 - B_1}{B_2} \times 35.5$

Question

An oxide of metal have 20% oxygen, then equivalent weight of oxide is:

- A** 32
- B** 48
- C** 40
- D** 52

Question

Sulphur trioxide is prepared by the following two reactions;



How many grams of SO_3 are produced from 1 mol of S_8 ?

Question

If specific heat of element is 3.2 cal/g. Find atomic mass of element?

- A** 1 g
- B** 2 g
- C** 3 g
- D** 4 g

Question

If the mass of neutron is doubled and that of proton is halved, the molecular mass of H_2O containing only H^1 and O^{16} atoms will

- A** increasing by about 25%
- B** decrease by about 25%
- C** increase by about 16.67%
- D** decrease by about 16.67%

Question

If Avogadro number N_A , is changed from $6.022 \times 10^{23} \text{ mol}^{-1}$ to $6.022 \times 10^{20} \text{ mol}^{-1}$ this would change: [NEET 2015]

- A** the definition of mass in units of grams
- B** the mass of one mole of carbon
- C** the ratio of chemical species to each other in a balanced equation
- D** the ratio of elements to each other in a compound

Question

How much of NaOH is required to neutralize 1500 cm^3 of 0.1 M HCl ?

- A** 40 g
- B** 4 g
- C** 6 g
- D** 60 g

Question

If 500 ml of 0.1 M HCl is added to 250 ml of 0.01 M NaOH the resulting solution is

- A** Acidic
- B** Basic
- C** Neutral
- D** None of these

Question

If 1 L of 0.1 M H_2SO_4 is added to 2 L of 0.1 M $\text{Ca}(\text{OH})_2$ the resulting solution is:

- ☐ A Acidic
- ☐ B Basic
- ☐ C Neutral
- ☐ D None of these

Question

What approximate volume of 0.40M Ba(OH)_2 must be added to 50.0 mL of 0.30 M NaOH to get a solution in which the molarity of the OH^- ions is 0.50 M ?

- A** 33 mL
- B** 66 mL
- C** 133 mL
- D** 100 mL

Question

50 mL of 0.5 M oxalic acid is needed to neutralize 25 ml of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is:

- A** 40 g
- B** 10 g
- C** 20 g
- D** 80 g